

EC442: Lecture Notes

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Notes on the 1st year Macro class

October 05, 2026

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1. Lecture 1 Sep. 29

1.1. Why heterogeneity

Complete markets + standard welfare assumptions:

- Competitive eq. Pareto efficient; planner representation w/ fixed weights $\alpha_i > 0$.
- Common discounting; preferences $u(c) - v(n)$. Consumption FOCs:

$$\alpha_i u'(c_t^i) = \alpha_j u'(c_t^j) \Rightarrow \frac{u'(c_t^i)}{u'(c_t^j)} = \frac{\alpha_j}{\alpha_i}. \quad (1.1)$$

Marginal-utility ratio constant across time/states.

- Common CRRA: $u'(c) = c^{-\sigma}$.

$$\frac{c_t^i}{c_t^j} = \left(\frac{\alpha_i}{\alpha_j} \right)^{\frac{1}{\sigma}}. \quad (1.2)$$

Fixed consumption shares $\rightarrow$ **perfect insurance** against idiosyncratic income shocks. Aggregate risk remains; unequal Pareto weights $\rightarrow$ unequal consumption.

- Strictly convex labor disutility; interior labor choice; aggregate conditions fixed: productivity $\uparrow \rightarrow$ labor $\uparrow$; consumption share unchanged.

Implications:

- Karl Marx
- Full insurance at odds w/ data

Lecture addition: Intuition for Complete Markets

- **Example:** 18 students; one high earner, others lower. Complete markets $\rightarrow$ high earner sells state-contingent claims $\rightarrow$ consumption decoupled from income.
- **Marx analogy:** "From each according to ability, to each according to needs." Efficient allocation equalizes marginal utilities regardless of income.
- **Empirical mismatch:** Data shows strong income-consumption correlation. Complete markets imply correlation vanishes $\rightarrow$ model unrealistic. Heterogeneity requires market incompleteness.

Recording: 00:13:54, 00:16:04

1.2. How to model incomplete markets?

1. **Exogenous:** assume some markets absent.
 1. e.g. save/borrow via noncontingent bonds; no state-contingent claims.
2. **Endogenous:** derive market incompleteness from frictions.
 1. e.g. limited commitment (participation constraints), private information.
 2. deeper but harder to do

Lecture addition: Course Roadmap

- **Methods:** Dynamic programming, numerical methods.
- **Background:** Equilibrium and welfare theorems.
- **Growth:** Basic models of economic growth.
- **Incomplete markets:** Endogenous (participation constraints/limited commitment), exogenous (asset structure), search models.

Recording: 00:31:25, 00:33:02

Lecture addition: Frictions for Endogenous Incompleteness

- **Limited commitment:** Contracts unenforceable. E.g., US car insurance limits; agents flee jurisdiction (e.g., to Frankfurt) to avoid liability → limits risk trading.
- **Private information:** Effort/talent unobservable. Income sharing → moral hazard (effort reduction) → complete insurance inefficient/impossible.

Recording: 00:22:10, 00:27:38

1.3. Planning problem

Set $n_t = A = 1$; $f(k) = F(k, 1)$. Given $k_0 \geq 0$. Discount factor $0 < \beta < 1$; depreciation $0 \leq \delta \leq 1$.

Choose sequences $c_t, k_{t+1} \geq 0$:

$$\max \sum_{t=0}^{\infty} \beta^t u(c_t) \text{ s.t. } c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t. \quad (1.3)$$

- **Lagrangian;** current-value multipliers λ_t :

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t [u(c_t) + \lambda_t (f(k_t) + (1 - \delta)k_t - c_t - k_{t+1})]. \quad (1.4)$$

- **Interior FOCs:**

$$\begin{aligned} c_t : u'(c_t) &= \lambda_t, \\ k_{t+1} : \lambda_t &= \beta \lambda_{t+1} [f'(k_{t+1}) + 1 - \delta]. \end{aligned} \quad (1.5)$$

Euler equation:

$$u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + 1 - \delta]. \quad (1.6)$$

- **Transversality condition:**

$$\lim_{T \rightarrow \infty} \beta^T \lambda_T k_{T+1} = \lim_{T \rightarrow \infty} \beta^T u'(c_T) k_{T+1} = 0. \quad (1.7)$$

Discounted shadow value of terminal capital $\rightarrow 0$.

- Increasing, concave u ; concave f ; finite discounted utility: feasibility + FOCs + TVC $\rightarrow$ global optimum. At corners, use KKT conditions.

1.4. Towards a recursive formulation

- Sequence approach: infinitely many dated choices + optimality conditions.
- Repeated trade-off: consume today vs. save for tomorrow.
- Stationary problem; current capital k_t summarizes relevant history.
- Recursion: continuation value summarizes future choices; same decision problem at every date.

Value function: maximal lifetime utility from initial capital k_0 :

$$v^*(k_0) = \max_{\{c_t, k_{t+1}\}_{t \geq 0}} \sum_{t=0}^{\infty} \beta^t u(c_t). \quad (1.8)$$

Feasible sequences: $c_t, k_{t+1} \geq 0$; $c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$, all $t \geq 0$.

Split date 0 from continuation:

$$v^*(k_0) = \max_{c_0, k_1} \left[u(c_0) + \beta \max_{\{c_t, k_{t+1}\}_{t \geq 1}} \sum_{t=1}^{\infty} \beta^{t-1} u(c_t) \right]. \quad (1.9)$$

- Outer max: $c_0, k_1 \geq 0$; $c_0 + k_1 = f(k_0) + (1 - \delta)k_0$.
- Inner max: given chosen k_1 , same feasibility constraints for all $t \geq 1$. Discounting restarts at date 1 $\rightarrow$ inner value $v^*(k_1)$.

Continuation value absorbs future choices. **Bellman equation** (current capital k , next-period capital k'):

$$v^*(k) = \max_{c, k'} [u(c) + \beta v^*(k')] \quad (1.10)$$

subject to

$$c + k' = f(k) + (1 - \delta)k, c, k' \geq 0. \quad (1.11)$$

Infinite sequence $\rightarrow$ one-period choice + discounted continuation value. Stationarity $\rightarrow$ same problem at every date.

Bellman equation = **functional equation**; unknown = function $v(k)$.

Questions:

- **Existence:** conditions for solution v ?
- **Uniqueness:** conditions for unique v within chosen function class?
- **Equivalence:** when does $v = v^*$, with maximizing policy solving original sequence problem?

Answer:

- *Recursive Macroeconomic Theory* (RMT), “Dynamic Programming” ([Ljungqvist & Sargent, 2018, ch. 3](#)).

Lecture addition: Computational Advantage of DP

- **Lagrangian**: Leads to 2nd-order difference eq. Solving via “shooting” → local, clunky.
- **Recursive**: Bellman eq. → direct computation of **global solution** via iteration.
- **Convergence**: VFI guaranteed to converge to unique fixed point (via CMT), unlike many ODE/PDE methods.

Recording: 01:06:37, 01:07:39

1.5. Dynamic Programming

Bellman operator T : candidate continuation value $v \rightarrow$ updated value Tv .

$$(Tv)(k) = \max_{c,k'} [u(c) + \beta v(k')] \quad (1.12)$$

subject to

$$c + k' = f(k) + (1 - \delta)k, c, k' \geq 0. \quad (1.13)$$

Bellman solution = **fixed point**:

$$Tv = v \Leftrightarrow (Tv)(k) = v(k) \text{ for all feasible } k. \quad (1.14)$$

Fixed-point theorems give existence/uniqueness conditions within chosen function space; assumptions matter.

1.6. Contraction mapping theorem

Theorem 1.6.1 (Contraction mapping theorem): Let (S, d) be a nonempty complete metric space and $T : S \rightarrow S$ a contraction: there exists $\beta \in (0, 1)$ such that

$$d(Tx, Ty) \leq \beta d(x, y) \text{ for all } x, y \in S. \quad (1.15)$$

Then T has a unique fixed point $v \in S$. For every $v_0 \in S$ and $n = 0, 1, 2, \dots$,

$$d(T^n v_0, v) \leq \beta^n d(v_0, v). \quad (1.16)$$

Hence $T^n v_0 \rightarrow v$ as $n \rightarrow \infty$, with geometric convergence. Here T^n denotes n successive applications of T , with T^0 the identity.

Lecture addition: Intuition for Contraction Mapping

- Analogous to Brouwer fixed-point theorem in 1D: continuous function mapping interval to itself with slope $< 1 \rightarrow$ unique fixed point.
- In function spaces, “slope” = contraction factor β . Iterating operator moves points closer $\rightarrow$ convergence to unique fixed point.

Recording: 00:59:06, 01:01:46

1.6.1. Application to planner

- Assume feasible capital remains in nonempty compact interval K .
- $S = C(K)$: continuous real-valued functions on K ; automatically bounded.
- **Sup-norm metric**:

$$d(v, w) = \|v - w\|_\infty = \sup_{k \in K} |v(k) - w(k)|. \quad (1.17)$$

(S, d) complete.

- Must verify $T : S \rightarrow S$: updated values remain bounded and continuous. Then Bellman discounting gives

$$\|Tv - Tw\|_\infty \leq \beta \|v - w\|_\infty. \quad (1.18)$$

Theorem 1.6.1 $\rightarrow$ unique fixed point in S ; value iteration converges from any $v_0 \in S$.

Lecture addition: Economic Interpretation of VFI

- **Initial guess** $v^0 = 0$: One-period model (no future utility). Agent consumes all output today.
- **First iteration** Tv^0 : Two-period model. Optimize today/tomorrow, world ends after tomorrow.
- **Convergence**: Each iteration extends horizon by 1 period. $n \rightarrow \infty \rightarrow$ infinite-horizon value function.

Recording: 00:56:48, 00:58:31

1.7. Using the value function analytically

Assume u , f , and v differentiable; optimal choices interior. Substitute $c = f(k) + (1 - \delta)k - k'$ into Bellman:

$$v(k) = \max_{k'} [u(f(k) + (1 - \delta)k - k') + \beta v(k')], \quad (1.19)$$

over feasible k' .

- **FOC for k'** :

$$-u'(c) + \beta v'(k') = 0 \Rightarrow u'(c) = \beta v'(k'). \quad (1.20)$$

- **Envelope condition**: differentiate w.r.t. k , holding optimal k' fixed:

$$v'(k) = u'(c)[f'(k) + 1 - \delta]. \quad (1.21)$$

- **Combine**: apply envelope condition next period $\rightarrow$ usual Euler equation:

$$u'(c_t) = \beta v'(k_{t+1}) = \beta u'(c_{t+1})[f'(k_{t+1}) + 1 - \delta]. \quad (1.22)$$

Same condition as sequence problem. At corners, use KKT conditions.

1.8. Using the value function numerically

Need numerical representation of $v(k)$.

- Simplest approach: restrict capital to finite grid

$$K_N = \{k^1, k^2, \dots, k^N\} \subset K. \quad (1.23)$$

- Approximate value function by vector

$$\mathbf{v} = (v_1, v_2, \dots, v_N)^\top \in \mathbb{R}^N, v_i \approx v(k^i). \quad (1.24)$$

One entry per capital grid point.

Algorithm 1.8.1 (Value function iteration (VFI)):

1. **Choose grid** $K_N = \{k^1, \dots, k^N\}$. For each pair (i, j) , compute

$$c_{ij} = f(k^i) + (1 - \delta)k^i - k^j. \quad (1.25)$$

Let J_i contain choices j with $c_{ij} \geq 0$ and finite $u(c_{ij})$. Each J_i must be nonempty.

2. **Initial guess:** any $\mathbf{v}^0 \in \mathbb{R}^N$, e.g. all zeros. Set iteration $m = 0$ and tolerance $\varepsilon > 0$.
3. **Bellman update:** for every $i = 1, \dots, N$,

$$v_i^{m+1} = \max_{j \in J_i} [u(c_{ij}) + \beta v_j^m]. \quad (1.26)$$

Use the old vector for all updates: $\mathbf{v}^{m+1} = T_N \mathbf{v}^m$.

4. **Convergence check:**

$$\Delta_m = \max_{1 \leq i \leq N} |v_i^{m+1} - v_i^m|. \quad (1.27)$$

If $\Delta_m \leq \varepsilon$, stop and return $\hat{\mathbf{v}} = \mathbf{v}^{m+1}$. Otherwise set $m \leftarrow m + 1$ and repeat step 3.

5. **Recover approximate policy:** for each i , choose

$$j_i \in \arg \max_{j \in J_i} [u(c_{ij}) + \beta \hat{v}_j]. \quad (1.28)$$

Set $k'(k^i) = k^{j_i}$ and $c(k^i) = c_{ij_i}$.

Why it converges: T_N is a β -contraction in the sup norm. **Theorem 1.6.1** $\rightarrow$ unique grid fixed point $\mathbf{v}_N^*$; **Algorithm 1.8.1** converges from any initial vector. At stopping, value-error bound:

$$\|\hat{\mathbf{v}} - \mathbf{v}_N^*\|_\infty \leq \frac{\beta}{1 - \beta} \Delta_m \leq \frac{\beta}{1 - \beta} \varepsilon. \quad (1.29)$$

This controls iteration error; grid approximation error remains.

Lecture addition: Convergence in General Equilibrium

- CMT guarantees convergence for individual decision problems (value functions).
- Heterogeneous GE: convergence of **distribution** of states (assets, income, human capital) required.
- Distribution = higher-dimensional object; may have multiple fixed points → more complex than single-agent VFI.

Recording: 01:14:16, 01:15:18

Lecture addition: Curse of Dimensionality

- Grid-based VFI suffers from **curse of dimensionality**: n state variables → M^n grid points (exponential growth).
- Large n (e.g., 50) → computationally intractable.
- **Frontier approaches**: Sparse grids; mean field games [ASR uncertain: Ben Moll]; heuristic trust in convergence when rigorous proofs unavailable.

Recording: 01:15:51, 01:16:53

Lecture addition: Complexity vs. Reality

- Model complexity driven by **expectation side**: agents anticipate others' actions → complex fixed-point problems.
- Evidence: real people not “fully forward-looking” as standard models assume.
- **Responses**: (1) Develop computational tools for complex models; (2) Simplify models to match cognitive complexity (behavioral/limited rationality).

Recording: 01:17:25, 01:19:00

2. Lecture 2 Oct 5

2.1. Recall

2.1.1. Bellman equation for the standard growth model

$$v(k) = \max_{c,k'} [u(c) + \beta v(k')], u(c) = \frac{c^{1-\sigma}}{1-\sigma}, \quad (2.1)$$

subject to

$$c + k' = (1 - \delta)k + k^\alpha, c > 0, k' \geq 0. \quad (2.2)$$

For $\sigma = 1$, use $u(c) = \log c$.

2.1.2. Contraction mapping theorem

On a nonempty complete metric space (S, d) , a contraction $T : S \rightarrow S$ with modulus $\beta \in (0, 1)$ has a unique fixed point v ([Theorem 1.6.1](#)):

$$\begin{aligned} d(Tx, Ty) &\leq \beta d(x, y), \\ d(T^n v_0, v) &\leq \beta^n d(v_0, v). \end{aligned} \quad (2.3)$$

Iteration from any $v_0 \in S$ converges geometrically to v .

2.2. Blackwell's sufficient conditions

Theorem 2.2.1 (Blackwell's sufficient conditions): Let $B(X)$ be the bounded real-valued functions on X , equipped with the sup norm, and let $T : B(X) \rightarrow B(X)$. Suppose:

1. **Monotonicity:** For all $f, g \in B(X)$,

$$f(x) \geq g(x) \text{ for all } x \in X \implies (Tf)(x) \geq (Tg)(x) \text{ for all } x \in X. \quad (2.4)$$

2. **Discounting:** There exists $\beta \in (0, 1)$ such that, for every $f \in B(X)$ and $a \geq 0$,

$$(T(f + a))(x) \leq (Tf)(x) + \beta a \text{ for all } x \in X. \quad (2.5)$$

Here $(f + a)(x) = f(x) + a$.

Then T is a contraction with modulus β :

$$\|Tf - Tg\|_\infty \leq \beta \|f - g\|_\infty. \quad (2.6)$$

2.2.1. Verification for the Bellman operator

Maximization preserves pointwise inequalities, so $f \geq g \implies Tf \geq Tg$. Adding a constant to continuation utility gives

$$(T(f + a))(k) = \max_{c,k'} [u(c) + \beta f(k')] + \beta a = (Tf)(k) + \beta a. \quad (2.7)$$

Both conditions hold; the self-map requirement $T : B(X) \rightarrow B(X)$ must also be verified.

2.3. Blackwell's conditions for the standard model

Let $\alpha, \delta, \beta \in (0, 1)$. Resources and consumption:

$$\begin{aligned} y(k) &= \underbrace{(1 - \delta)k}_{\text{remaining capital}} + \underbrace{k^\alpha}_{\text{output}}, \\ c(k, k') &= y(k) - k'. \end{aligned} \quad (2.8)$$

The invariant state space satisfies

$$\begin{aligned} \bar{k} &= \delta^{-\frac{1}{1-\alpha}}, \\ X &= [0, \bar{k}], \\ 0 \leq y(k) &\leq y(\bar{k}) = \bar{k} \text{ for all } k \in X. \end{aligned} \quad (2.9)$$

Feasible savings and the Bellman operator:

$$\begin{aligned} \Gamma(k) &= \{k' \geq 0 : c(k, k') \geq 0\} \subseteq X, \\ (Tf)(k) &= \sup_{k' \in \Gamma(k)} \left[\underbrace{u(c(k, k'))}_{\text{current utility}} + \beta \underbrace{f(k')}_{\text{continuation value}} \right]. \end{aligned} \quad (2.10)$$

Extend u to $c = 0$ by its limit, possibly $-\infty$. Replace $\sup$ with $\max$ when the optimum is attained.

2.3.1. Monotonicity

For $f, g \in B(X)$ with $f \geq g$ pointwise,

$$\begin{aligned} (Tf)(k) &= \sup_{k' \in \Gamma(k)} [u(c(k, k')) + \beta f(k')] \\ &\geq \sup_{k' \in \Gamma(k)} [u(c(k, k')) + \beta g(k')] \\ &= (Tg)(k). \end{aligned} \quad (2.11)$$

2.3.2. Discounting

For $f \in B(X)$ and a constant $a \geq 0$,

$$\begin{aligned} (T(f + a))(k) &= \sup_{k' \in \Gamma(k)} \left[u(c(k, k')) + \beta f(k') + \underbrace{\beta a}_{\text{constant in } k'} \right] \\ &= (Tf)(k) + \beta a. \end{aligned} \quad (2.12)$$

2.3.3. Boundedness and conclusion

For $0 < \sigma < 1$, $u(0) = 0$ and

$$\begin{aligned} 0 \leq u(c(k, k')) &\leq u(\bar{k}) < \infty, \\ \|Tf\|_\infty &\leq u(\bar{k}) + \beta \|f\|_\infty < \infty. \end{aligned} \quad (2.13)$$

Thus $T : B(X) \rightarrow B(X)$ on the complete sup-norm space $B(X)$. By [Theorem 2.2.1](#) and [Theorem 1.6.1](#), there is a unique $v \in B(X)$ with $Tv = v$, and for every $v_0 \in B(X)$,

$$\|T^n v_0 - v\|_\infty \leq \beta^n \|v_0 - v\|_\infty \rightarrow 0. \quad (2.14)$$

For $\sigma \geq 1$ (including log utility),

$$k = 0 \implies \Gamma(k) = \{0\} \implies (Tf)(0) = -\infty. \quad (2.15)$$

The bounded-function theorem then requires a different state or function space.

2.4. Formulating recursive models

Identify the state variables before writing the Bellman equation.

A **state variable**:

- Matters for the decision problem.
- Is taken as given by the decision maker at the time of choice.
- Varies over time or across states.

Other variables:

- **Choice variables**: chosen by the decision maker.
- **Parameters**: affect the problem but are fixed over time and across states.

The value function takes the state variables as its arguments.

2.4.1. How to determine the state

1. **Fix timing**: Specify what is observed before choices are made. A lagged choice may be today's state; a newly chosen quantity is a control.
2. **List candidates**: Retain inherited stocks and observed shocks that affect current constraints, utility, or forecasts. Separate fixed parameters.
3. **Check sufficiency**: Given state s and choices q , specify

$$\begin{aligned} q &\in \Gamma(s), \\ \text{current utility} &= U(s, q), \\ s' &= G(s, q, \varepsilon'). \end{aligned} \quad (2.16)$$

Here ε' is future uncertainty. Two histories with the same s must give the same feasible choices, current utility, and conditional distribution of s' for every feasible q .

4. **Remove redundancy**: Drop a coordinate only if this sufficiency test still holds. A one-to-one change of coordinates gives an equivalent state.
5. **Write the Bellman equation**:

$$v(s) = \sup_{q \in \Gamma(s)} \left[\underbrace{U(s, q)}_{\text{current utility}} + \beta \underbrace{\mathbb{E}[v(s') | s, q]}_{\text{expected continuation}} \right]. \quad (2.17)$$

Use max when the optimum is attained; omit the expectation in a deterministic model.

Reduction test: If two candidate states have the same proposed summary but different feasible choices, utility, or transition distributions, the summary is insufficient.

2.4.2. Example: standard model with two sectors

Each period, allocate inherited capital k and one unit of labor freely between consumption (c) and investment (i) sectors. Let $\alpha, \gamma \in (0, 1)$.

Feasibility:

$$\begin{aligned}
 c &= \underbrace{k_c^\alpha n_c^{1-\alpha}}_{\text{consumption output}}, \\
 k' &= \underbrace{(1-\delta)k}_{\text{remaining capital}} + \underbrace{k_i^\gamma n_i^{1-\gamma}}_{\text{investment output}}, \\
 k_c + k_i &= k, \\
 n_c + n_i &= 1.
 \end{aligned} \tag{2.18}$$

All quantities are nonnegative.

State identification:

- k is inherited; (k_c, k_i, n_c, n_i) are chosen before production.
- Free reallocation removes dependence on the previous sectoral split.
- Given k and current choices, utility and next-period k' are determined.

Hence $s = k$ is sufficient.

The Bellman equation is

$$v(k) = \max_{c, k', k_c, k_i, n_c, n_i} [u(c) + \beta v(k')], \tag{2.19}$$

subject to the constraints above, with u as defined in Recall.

2.4.3. Example: two sectors with irreversible investment

Capital is installed before the period and cannot move between sectors. Choose labor allocations and new investment x_c, x_i today; investment becomes productive next period.

Feasibility:

$$\begin{aligned}
 c &= k_c^\alpha n_c^{1-\alpha}, \\
 x_c + x_i &= \underbrace{k_i^\gamma n_i^{1-\gamma}}_{\text{investment output}}, \\
 k'_c &= (1-\delta)k_c + x_c, \\
 k'_i &= (1-\delta)k_i + x_i, \\
 n_c + n_i &= 1, \\
 x_c, x_i &\geq 0.
 \end{aligned} \tag{2.20}$$

All other quantities are nonnegative. Irreversibility implies

$$k'_j \geq (1-\delta)k_j \text{ for } j \in \{c, i\}. \tag{2.21}$$

Sectoral capital can decline only through depreciation.

2.4.4. Standard model with shocks

Productivity z is i.i.d. over time. The resource constraint is

$$c + k' = (1 - \delta)k + zk^\alpha. \quad (2.22)$$

In all formulations, $c, k' \geq 0$.

2.4.4.1. States (k, z) : after the shock

Observe z , then choose c and k' :

$$v(k, z) = \max_{c, k'} [u(c) + \beta \mathbb{E}_{z'} [v(k', z')]], \quad (2.23)$$

subject to Equation (2.22). The expectation is over next period's shock z' .

2.4.4.2. State x : total resources

After observing z , define

$$x = (1 - \delta)k + zk^\alpha. \quad (2.24)$$

Then

$$w(x) = \max_{c, k'} [u(c) + \beta \mathbb{E}_{z'} [w(x')]], \quad (2.25)$$

subject to

$$\begin{aligned} c + k' &= x, \\ x' &= (1 - \delta)k' + z'(k')^\alpha. \end{aligned} \quad (2.26)$$

z' is independent of z , so the transition of x' requires only k' .

Note: This reduction to x relies on z being i.i.d. over time.

2.4.4.3. State k : before the shock

$V(k)$ is the value *before* z is observed. Choices $c(z)$ and $k'(z)$ may *depend on its realization*:

Choices. For a fixed (k, z) or x , Options 1 and 2 maximize over scalar choices c, k' . Their policy functions are $c(k, z), k'(k, z)$ or $c(x), k'(x)$; state arguments are implicit. Option 3 maximizes over functions $c(z), k'(z)$ for every possible z .

$$V(k) = \max_{c(z), k'(z)} \mathbb{E}_z [u(c(z)) + \beta V(k'(z))], \quad (2.27)$$

subject to

$$c(z) + k'(z) = (1 - \delta)k + zk^\alpha \text{ for every } z. \quad (2.28)$$

$$\begin{aligned} v(k, z) &= w((1 - \delta)k + zk^\alpha), \\ V(k) &= \mathbb{E}_z [v(k, z)]. \end{aligned} \quad (2.29)$$

Timing. Before observing z , $V(k)$ averages over its realization; i.i.d. shocks make past productivity uninformative about z . After observing z , the state records realized resources through (k, z) or x . Decisions can depend on observed z in all three formulations.

2.5. Equilibrium

A social planner maximizes welfare subject to feasibility constraints. In a decentralized economy, agents make their own choices, linked through markets, institutions, and policies.

- How can decentralized outcomes be represented formally?
- How do equilibrium allocations relate to the planner's solution?
- Can decentralized equilibria be formulated recursively?

2.6. What is a competitive equilibrium?

A competitive equilibrium consists of:

- An *allocation*: consumption, production, and factor inputs.
- A *price system*: prices in every market.

Two conditions:

- **Optimization:** Households maximize utility and firms maximize profits, taking prices as given.
- **Feasibility:** The allocation is feasible: total use of each good does not exceed production plus endowments.

2.7. Sequential equilibrium

In the deterministic standard model, markets open each period. The household owns capital k_t and a fixed labor endowment $n_t = 1$. It rents capital and supplies labor to the firm; both agents take prices as given. Let r_t be the net return on capital. The capital rental rate is $r_t + \delta$; the wage is w_t .

Given k_0 , a sequential equilibrium consists of an allocation $\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}$ and prices $\{r_t, w_t\}_{t=0}^{\infty}$ satisfying the following conditions.

2.7.1. Household

Given the price paths, the household solves

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t), \quad (2.30)$$

subject to

$$\begin{aligned} c_t + k_{t+1} &= \underbrace{(1 + r_t)k_t}_{\text{capital payoff}} + \underbrace{w_t n_t}_{\text{labor income}}, \\ n_t &= 1, \\ c_t &\geq 0, \\ k_{t+1} &\geq 0. \end{aligned} \quad (2.31)$$

2.7.2. Firm

The firm chooses capital demand K_t and labor demand N_t each period:

$$\max_{K_t, N_t \geq 0} [K_t^\alpha N_t^{1-\alpha} - (r_t + \delta)K_t - w_t N_t]. \quad (2.32)$$

2.7.3. Market clearing

$$\begin{aligned} K_t &= k_t, \\ N_t &= n_t = 1, \\ c_t + k_{t+1} &= (1 - \delta)k_t + k_t^\alpha. \end{aligned} \quad (2.33)$$

At positive inputs, the firm's first-order conditions imply

$$\begin{aligned} r_t &= \alpha k_t^{\alpha-1} - \delta, \\ w_t &= (1 - \alpha)k_t^\alpha. \end{aligned} \quad (2.34)$$

2.7.4. No-Ponzi condition

If borrowing is allowed, replace $k_{t+1} \geq 0$ with

$$\liminf_{t \rightarrow \infty} \frac{k_{t+1}}{\prod_{s=1}^t (1 + r_s)} \geq 0. \quad (2.35)$$

At an interior optimum with $u'(c_s) > 0$, the Euler equation gives

$$1 + r_s = \frac{u'(c_{s-1})}{\beta u'(c_s)}. \quad (2.36)$$

Multiplying from $s = 1$ to t , marginal utilities cancel:

$$\begin{aligned} \prod_{s=1}^t (1 + r_s) &= \prod_{s=1}^t \frac{u'(c_{s-1})}{\beta u'(c_s)}, \\ &= \frac{u'(c_0)}{\beta^t u'(c_t)}. \end{aligned} \quad (2.37)$$

Hence

$$\frac{k_{t+1}}{\prod_{s=1}^t (1 + r_s)} = \frac{\beta^t u'(c_t) k_{t+1}}{u'(c_0)}. \quad (2.38)$$

Since $u'(c_0) > 0$ is constant, dividing by it preserves the sign of the limit inferior. Thus the no-Ponzi condition is equivalent to

$$\liminf_{t \rightarrow \infty} \beta^t u'(c_t) k_{t+1} \geq 0. \quad (2.39)$$

The household cannot finance consumption by rolling over debt forever.

2.8. Arrow-Debreu equilibrium

Sequential markets open each period. In an Arrow-Debreu equilibrium, markets open once at date 0. Agents contract for goods delivered at each future date (and state, under uncertainty). Delivery and production still occur each period.

Given k_0 , an Arrow-Debreu equilibrium consists of an allocation $\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}$ and prices $\{p_t, r_t, w_t\}_{t=0}^{\infty}$ satisfying the conditions below.

The additional price $p_t > 0$ is the date-0 price of one unit of the good delivered at date t , with $p_0 = 1$. It converts date- t quantities into date-0 goods. The wage w_t and capital rental rate $r_t + \delta$ remain measured in date- t goods. Their date-0 prices are $p_t w_t$ and $p_t(r_t + \delta)$.

2.8.1. Household

The household solves

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t), \quad (2.40)$$

subject to $n_t = 1$, $c_t \geq 0$, given k_0 , and a single present-value budget:

$$\sum_{t=0}^{\infty} p_t(c_t + k_{t+1}) = \sum_{t=0}^{\infty} p_t[(1 + r_t)k_t + w_t n_t]. \quad (2.41)$$

The present-value sums are assumed finite.

2.8.2. Firm and market clearing

For each delivery date t , the firm solves

$$\max_{K_t, N_t \geq 0} [K_t^\alpha N_t^{1-\alpha} - (r_t + \delta)K_t - w_t N_t]. \quad (2.42)$$

Multiplying profits by $p_t > 0$ leaves the input choices unchanged. Markets clear at every date:

$$\begin{aligned} K_t &= k_t, \\ N_t &= n_t = 1, \\ c_t + k_{t+1} &= (1 - \delta)k_t + k_t^\alpha. \end{aligned} \quad (2.43)$$

2.8.3. No arbitrage and the present-value budget

In the household's budget, one unit of k_{t+1} costs p_t and generates a payoff worth $p_{t+1}(1 + r_{t+1})$ at date 0. No arbitrage requires

$$\begin{aligned} p_t - p_{t+1}(1 + r_{t+1}) &= 0, \\ \frac{p_t}{p_{t+1}} &= 1 + r_{t+1}. \end{aligned} \quad (2.44)$$

Thus $\frac{p_t}{p_{t+1}}$ is the gross real return from t to $t + 1$. With $p_0 = 1$,

$$p_t = \frac{1}{\prod_{s=1}^t (1 + r_s)}. \quad (2.45)$$

Capital purchases and their future payoffs cancel from the lifetime budget:

$$\sum_{t=0}^{\infty} p_t c_t = \underbrace{(1 + r_0)k_0}_{\text{initial capital wealth}} + \underbrace{\sum_{t=0}^{\infty} p_t w_t}_{\text{lifetime labor income}} . \quad (2.46)$$

No separate no-Ponzi condition is needed: the date-0 budget already limits lifetime spending to lifetime wealth.

Bibliography

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